INCH-POUND
MIL-DTL-62576B
30 September 1998
SUPERSEDING
MIL-PRF-62576A
13 March 1997

DETAIL SPECIFICATION

REGULATOR, ENGINE GENERATOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a solid state regulator utilizing electronic devices to regulate the voltage and current of the 28 volt (V), direct current (dc) generator (see 6.1).

1.2 Classification. The solid state regulators are of the types listed below:

Type I	- Regulator (Drawing 11631857), to operate with 28 V, 25 ampere (A), dc generator.
Type II	- Regulator, Single Current Limit (Drawing 12257823), to operate with 28 V, 300 A, dc generator.
Type III	- Regulator, Dual Current Limit (Drawing 11672403), to operate with 28 V, 300 A, dc generator.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 2920

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:50Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirement documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

DEPARTMENT OF DEFENSE

MIL-B-11188	- Batteries, Storage: Lead-Acid.
MIL-PRF-62061	- Generator, Engine, Accessory, 28 Volt DC, Rated Output 300 Amps.
MIL-B-62346	- Battery, Storage: Lead-Acid (Low Maintenance) Type 6TL.

STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-202	- Test Methods for Electronic and Electrical Component Parts.
MIL-STD-462	- Measurement of Electromagnetic Interference Characteristics, Test Method for.
MIL-STD-1275	- Characteristics of 28 V dc Electrical Systems in Military Vehicles.
MS39061	- Switch, Toggle - SPST, SPDT, 24 Volt dc, 25 Amp (Waterproof).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:50Z
Check the source to verify that this is the current version before use.

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

NUCLEAR REGULATORY COMMISSION (NRC)

Code of Federal Regulations (CFR) - Title 10, Parts 30 and 40.

(Copies of the Code of Federal Regulations (CFR) are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

DRAWINGS

ARMY

7355736	- Generator, 28 V dc, 25 A.
10950808	- Generator, 28 V dc, 25 A.
11631857	- Regulator, Engine Generator, 28 V dc 25 A, Solid State.
11672403	- Voltage Regulator, Dual Current Limit, Solid State.
12257823	- DC Solid State Regulator.

(Copies of these drawings are available from the U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/SAE AS478 - Identification Marking Methods (DoD Adopted).

(Application for copies should be addressed to American National Standards Institute, 11 West 42nd Street, New York, NY 10036.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A123 - Standard Specification for Zinc (Hop Dip Galvanized) Coatings on Iron and Steel Products (AASHTO M111) (DoD Adopted).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:50Z
Check the source to verify that this is the current version before use.

ASTM A653	- Standard Specification for Sheet Steel, Zinc Coated (Galvanized) (Galvanealed) by the Hot Dip Process.
ASTM A924	- Standard Specification for General Requirements for Sheet Steel, Metallic-Coated by the Hot Dip.
ASTM B117	- Standard Practice for Operating Salt Spray (Fog) Apparatus (DoD Adopted).
ASTM G21	- Standard Practice for Determining Resistance of Synthetic Polymeric Materials to Fungi (DoD Adopted).

(Application for copies of ASTM publications may be obtained from the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

GENERAL MOTORS CORPORATION

GM 9540P	- Accelerated Corrosion Test.

(Copies of GM publications may be obtained from General Motors Corporation, c/o Global Engineering, 15 Inverness Way, Englewood, CO 80112.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Materials. Materials shall be as specified herein, and on applicable drawings, standards and specifications. When the material used is not specifically covered in any of the referenced documents, it shall withstand the temperature range encountered within the component for which it is intended for use. There shall be no change in physical or chemical properties resulting in calibration or operation that fall outside the limits specified herein (see 4.4.3).

3.2.1 Dissimilar metals. Except where necessary to complete an electrical circuit, contact between dissimilar metals, which would encourage galvanic action, shall be avoided.

3.2.2 Finishes. All exterior surfaces of the regulator, except for the electrical connections and mating surfaces on the mounting bracket, shall be cleaned, treated, and painted (see 4.4).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:50Z
Check the source to verify that this is the current version before use.

3.2.3 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.2.3.1 Hazardous materials. Asbestos, cadmium, and radioactive material shall not be used in this item. Radioactive material is defined by CFR, Title 10, Parts 30 and 40, and other radioactive material in which the specific activity is greater than 0.002 microcuries per gram or the activity per item equals or exceeds 0.01 microcuries.

3.3 Design and construction.

3.3.1 Configuration. Configuration construction shall conform to applicable drawings specified herein (see 6.2).

3.3.2 Type I. The regulator shall be fabricated and assembled in accordance with Drawing 11631857 (see 4.4 and 4.4.3).

3.3.3 Type II. The regulator shall be fabricated and assembled in accordance with Drawing 12257823 (see 4.4).

3.3.4 Type III. The regulator shall be fabricated and assembled in accordance with Drawing 11672403 (see 4.4).

3.3.5 Threaded surfaces. The thread form, class and the number of threads per inch shall be as specified on the applicable drawings (see 4.4).

3.3.6 Mounting position. The regulator shall be designed to operate when mounted in any position and without the use of shock absorbing material or equipment (see 4.4).

3.4 Operating requirements.

3.4.1 Polarity. The regulator shall be designed to operate with a negative-grounded, electrical system and shall be internally case grounded (see 4.4).

3.4.2 Reverse current (I_R). The reverse current shall be as specified in 3.4.2.1 and 3.4.2.2 when measured between regulator and batteries (see 4.5.1).

3.4.2.1 Type I. I_R shall be 5 milliamperes (mA) maximum.

3.4.2.2 Types II and III. I_R shall be 30 mA maximum. For type III only, this requirement shall be met with pin “E” of the battery connector grounded and ungrounded.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:50Z
Check the source to verify that this is the current version before use.

3.4.3 Transient voltage.

3.4.3.1 Type I. The regulator shall meet the abnormal system requirements of MIL-STD-1275, except that the steady state voltage shall be no greater than 30 V and the transient voltage shall be no greater than 80 V (see 4.5.2.1 and 4.5.4).

3.4.3.2 Types II and III. The regulator shall meet the requirements of MIL-STD-1275 (see 4.5.2.2). For type III only, this requirement shall be met with pin "E" of the battery connector grounded and ungrounded.

3.4.4 Load switching.

3.4.4.1 Type I. The regulator shall meet the abnormal system requirements of MIL-STD-1275, except that the steady state voltage shall be no greater than 30 V, the transient voltage shall be no greater than 80 V, and the peak to peak ripple voltage shall be no greater than 5 V (see 4.5.3.1).

3.4.4.2 Types II and III. Not applicable.

3.4.5 Voltage regulation (E_R).

3.4.5.1 Type I. The regulator shall establish and maintain output voltage at 28.5 $\pm$ 0.7 V throughout the generator speed range of 2000 to 8000 revolutions per minute (rpm) (see 4.5.4.1).

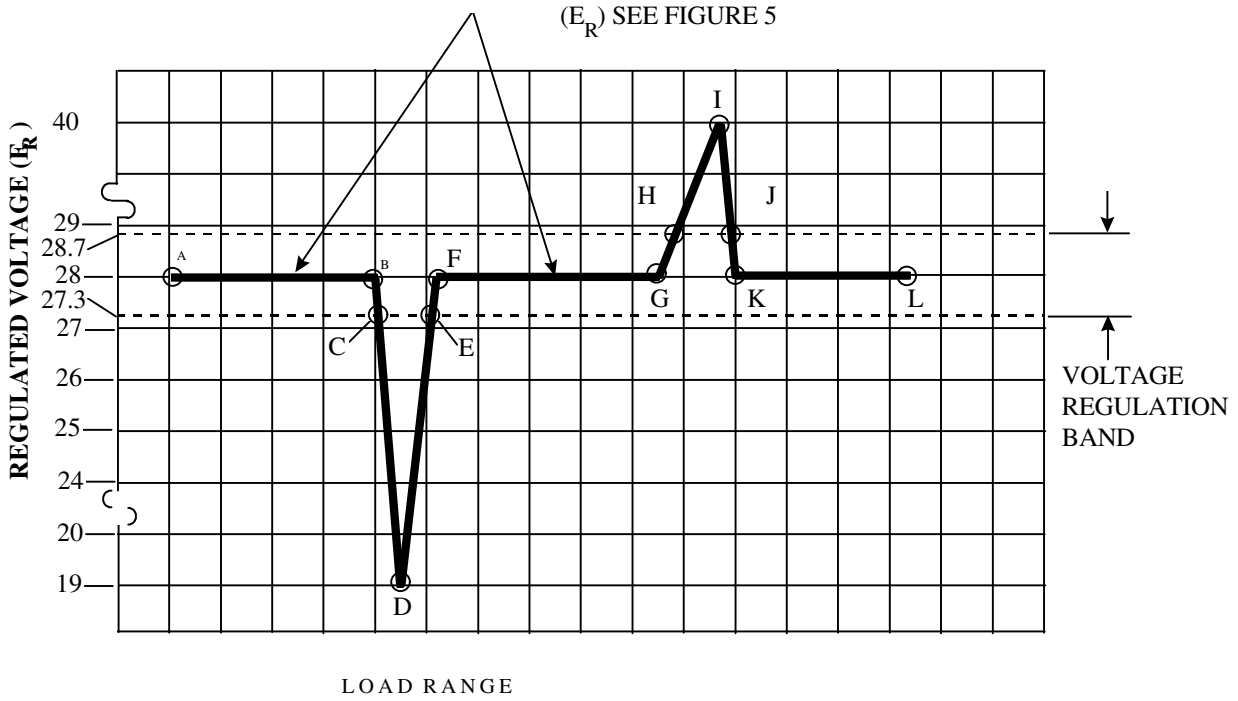
3.4.5.2 Types II and III. The regulator shall establish and maintain regulated voltage over a load range of 25 to 350 A and generator speed range of 2400 to 8000 rpm as shown in figures 1 and 2. Overshoot and undershoot limits and time to re-establish and maintain regulated voltage shall conform to limits on figure 1. The fundamental frequency of the voltage waveforms shall be greater than 100 Hertz (Hz) (see 4.5.4.2 and 4.5.4.3). For type III only, this requirement shall be met with pin "E" of the battery connector ungrounded, and when grounded, the 350 A requirement is replaced with 100 A.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:50Z
Check the source to verify that this is the current version before use.

3.4.6 Voltage ripple.

3.4.6.1 Type I. Not applicable.

3.4.6.2 Types II and III. The regulator output voltage waveform shall have the following limitations. The ac peak to peak ripple voltage shall be no more than 7 V. With batteries connected, the peak to peak ripple voltage shall be no greater than 4 V, except that excursions of less than 1 millisecond (ms) may exceed 4 V (see 4.5.5). For type III only, this requirement shall be met with pin "E" of the battery connector grounded and ungrounded.



(A-B) - BATTERY LOAD ONLY PRIOR TO LOAD APPLICATION

(B) - POINT OF LOAD APPLICATION

(C-E) - VOLTAGE UNDERSHOOT RECOVERY TIME

(D) - UNDERSHOOT VOLTAGE

(F-G) - REGULATED VOLTAGE (ER) DURING APPLIED LOAD (STEADY STATE)

(G) - POINT OF LOAD DISCONNECTION

(H-J) - VOLTAGE OVERSHOOT RECOVERY TIME

(I) - OVERSHOOT VOLTAGE

(K-L) - BATTERY VOLTAGE AFTER LOAD DISCONNECT

(C-E) - MAX. TIME LIMIT - 0.125 SEC

(H-J) - MAX. TIME LIMIT - 0.300 SEC

FIGURE 1. Voltage regulation limits

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:50Z
Check the source to verify that this is the current version before use.

3.4.7 Current regulation.

3.4.7.1 Type I. With generator operating at 2400 to 8000 rpm, the regulator shall maintain output current of 25 A at 28.5 ± 1.5 V (see 4.5.6.1).

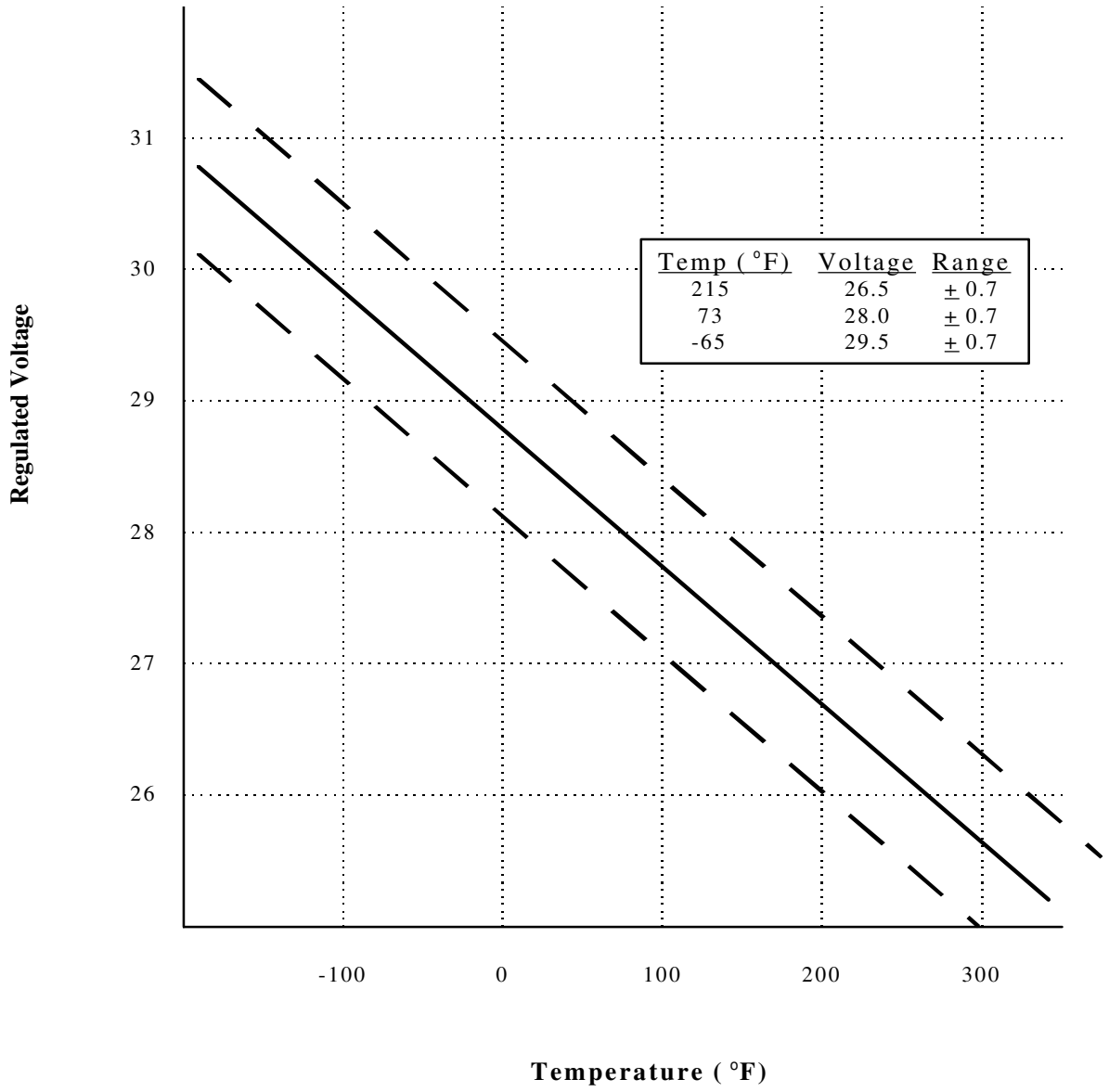


FIGURE 2. Temperature compensated voltage.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:50Z
Check the source to verify that this is the current version before use.

3.4.7.2 Types II and III (pin "E" ungrounded). With the generator operating over the speed range of 2400 to 8000 rpm, the regulator shall provide the following control (see 4.5.6.2 and 4.5.6.3):

a. No less than 350 A shall be maintained within the voltage regulation range of 28 ± 0.7 V.
b. The regulator shall limit the generator output current within the limits shown in figure 3.
c. The regulator shall attain stable current regulation in the current limit mode within 0.3 seconds of load application.
d. The regulator shall attain stable voltage regulation upon load release within 0.3 seconds after entering the lower voltage regulation limit (27.3 V) shown in figure 4.
e. In the current limit mode, the frequency of oscillation of the dc voltage shall have a period of less than 14 ms (T as shown in figure 4) and a peak-to-peak value (V as shown in figure 4) of less 10 percent (%) of the dc voltage.

3.4.7.3 Type III (pin "E" grounded). With the generator operating at 3000 rpm, the regulator shall provide the following control (see 4.5.6.4 and 4.5.6.5):

a. No less than 100 A shall be maintained within the voltage regulation range of 28 ± 0.7 V.
b. The regulator shall limit the generator output current within the limits shown in figure 3.
c. The regulator shall attain stable current regulation in the current limit mode within 0.3 seconds of load application.
d. The regulator shall attain stable voltage regulation upon load release within 0.3 seconds after entering the lower voltage regulation limit (27.3 V) shown in figure 4.
e. In the current limit mode, the frequency of oscillation of the dc voltage shall have a period of less than 14 ms (T as shown in figure 4) and a peak-to-peak value (V as shown in figure 4) of less 10 percent (%) of the dc voltage.

3.4.8 Overvoltage.

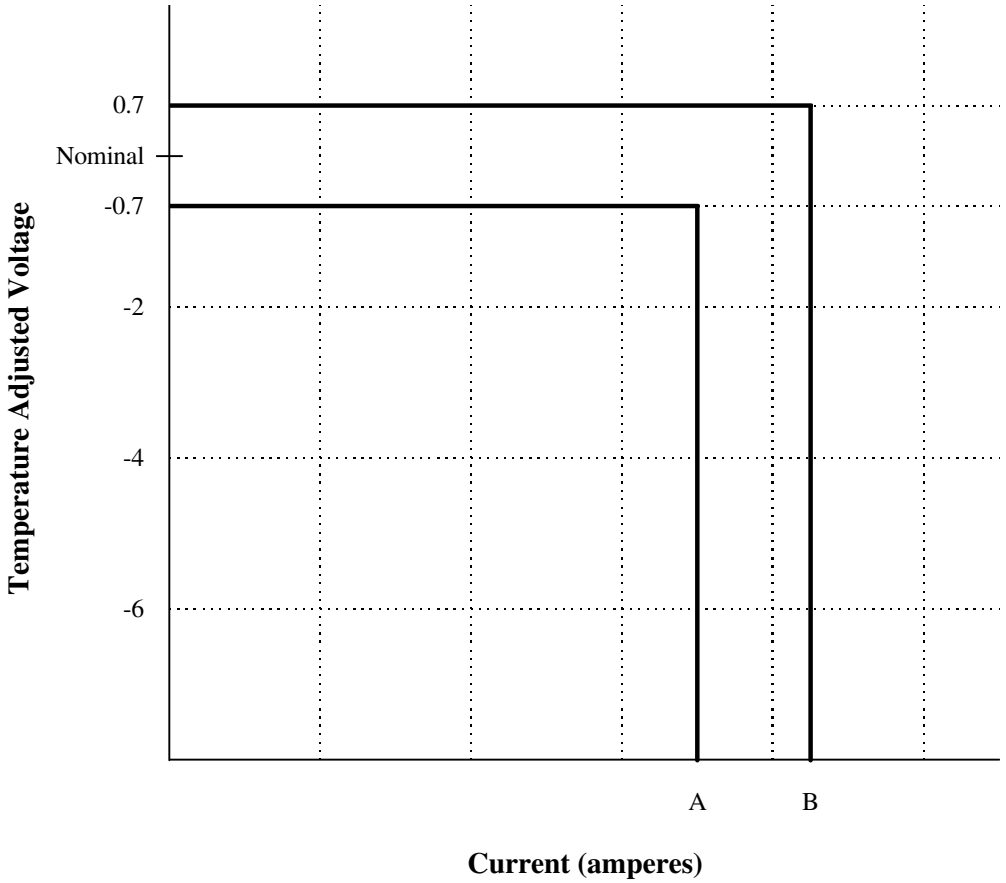
3.4.8.1 Type I. Not applicable.

3.4.8.2 Types II and III. The regulator generator system shall be protected from any failure in the regulator which will cause the regulator to go "full on". The overvoltage portion of the regulator shall react to deactivate the generator when the system voltage reaches a value of 33 ± 1 V. Overvoltage reaction time shall be not less than 0.25 second and not more than 1 second. Reset shall be manual and located on the regulator (see 4.5.7.1 and 4.5.7.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:50Z
Check the source to verify that this is the current version before use.

For type III only, this requirement shall be met with pin "E" of the battery connector grounded and ungrounded.

3.4.9 Field excitation. The regulator shall be activated and boot-strapped to regulation when the generator residual voltage reaches the value specified in 3.4.9.1 and 3.4.9.2 (see 4.5.8).



	Type II	Type III (pin "E" ungrounded)	Type III (pin "E" grounded)
Point A	350 A	350 A	100 A
Point B	420 A	420 A	180 A

FIGURE 3. Current limit curve, types II and III regulators.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:50Z
Check the source to verify that this is the current version before use.

3.4.9.1 Type I. The residual voltage shall be no more than 1.5 V.

3.4.9.2 Types II and III. The residual voltage shall be no less than 0.7 V. For type III only, this requirement shall be met with pin "E" of the battery connector grounded and ungrounded.

3.4.10 Effective field circuit resistance.

3.4.10.1 Type I. The regulator resistance at all times shall be no greater than 2 ohms (Ω) (see 4.5.9).

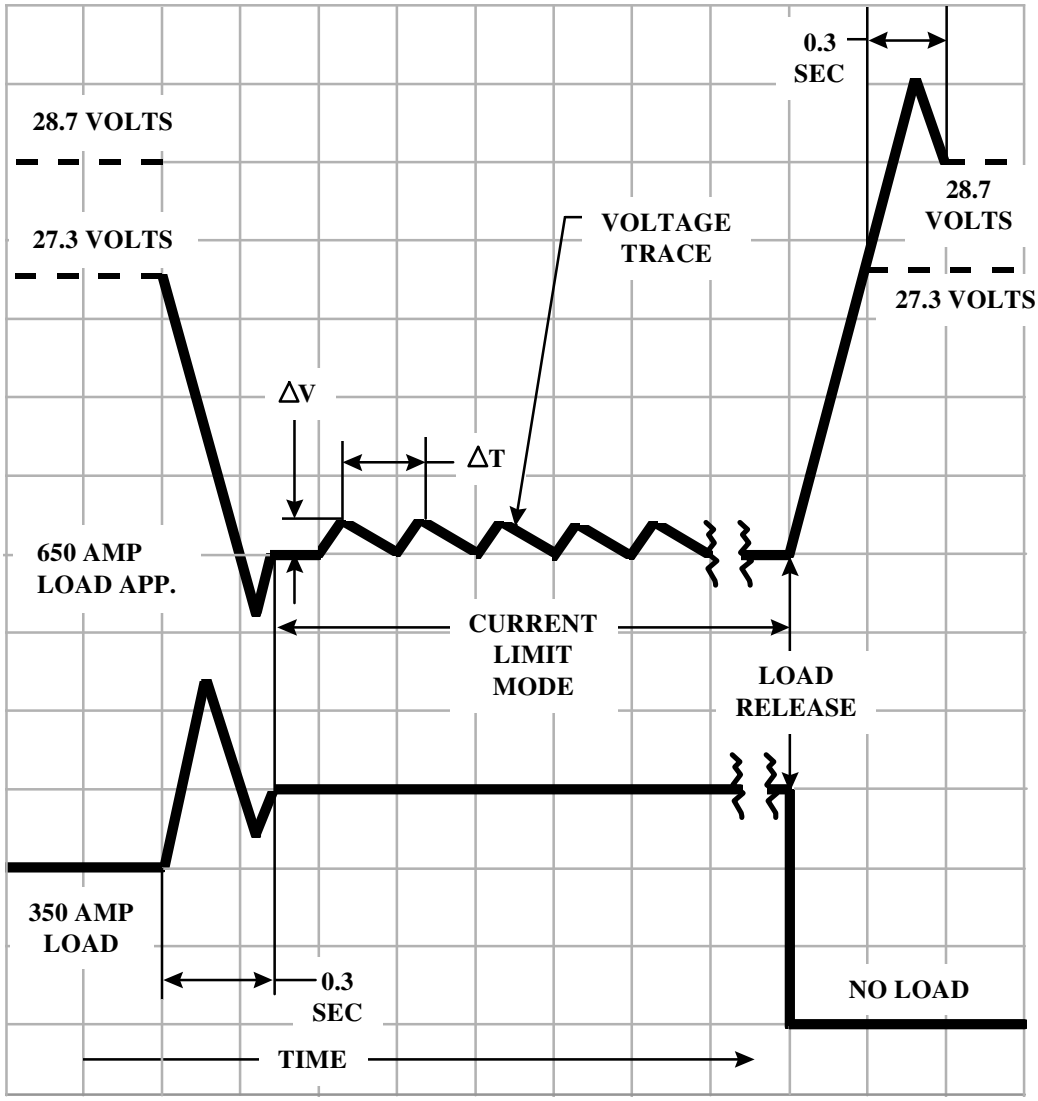


FIGURE 4. Current and voltage recovery time.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:50Z
Check the source to verify that this is the current version before use.

3.4.10.2 Types II and III. Not applicable.

3.4.11 Negative voltage operation.

3.4.11.1 Type I. Not applicable.

3.4.11.2 Types II and III. The regulator shall not be damaged by the application of negative voltage (see 4.5.10). For type III only, this requirement shall be met with pin "E" of the battery connector grounded and ungrounded.

3.4.12 Field circuit.

3.4.12.1 Type I. Not applicable.

3.4.12.2 Types II and III. The regulator shall be protected against any inductive energy that may be applied as a result of opening switch S_1 shown in figure 5 (see 4.5.11). For type III only, this requirement shall be met with pin "E" of the battery connector grounded and ungrounded.

3.4.13 Reverse battery polarity.

3.4.13.1 Type I. The regulator shall protect the generating system against damage by opening the circuit and shall prevent reversal of generator polarity (see 4.5.12).

3.4.13.2 Type II and III. Not applicable.

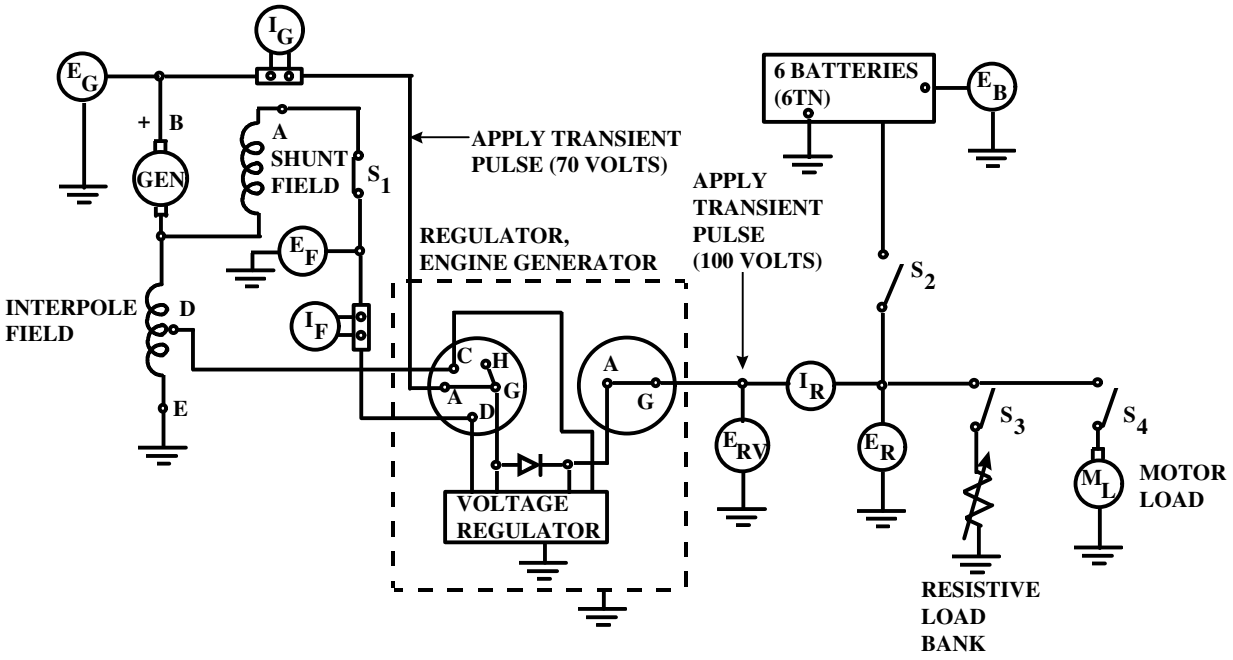
3.4.14 Short circuit protection.

3.4.14.1 Type I. Not applicable.

3.4.14.2 Types II. Regulator circuitry shall be provided to prevent damage to internal components in the event of a short circuit from pin "D" or "H" of the generator connector to ground during normal regulator operations (see 4.5.13).

3.4.14.3 Type III. Regulator circuitry shall be provided to prevent damage to internal components in the event of a short circuit from pin "D", "F", or "H" of the generator connector to ground during normal regulator operations (see 4.5.13). This requirement shall be met with pin "E" of the battery connector grounded and ungrounded.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:50Z
Check the source to verify that this is the current version before use.



TEST READINGS REQUIRED

I_G - GEN AMPS	E_B - BATTERY VOLTS	S_1 - SWITCH USED DURING FORDING MS39061-1 OR EQUAL
E_G - GEN VOLTS	E_R - REGULATED VOLTAGE	
E_F - GEN FIELD VOLTS	E_{RV} - RIPPLE VOLTAGE	R_L - RESISTIVE LOAD BANK
I_F - GEN FIELD AMPS	I_R - REVERSE CURRENT	

NOTE:

For type III only, for all tests with battery connector pin “E” grounded, generator Connector pin “D” shall be replaced with pin “F”.

FIGURE 5. Test circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:50Z
Check the source to verify that this is the current version before use.

3.4.15 Generator accessory sensing circuit.

3.4.15.1 Type I. Not applicable.

3.4.15.2 Types II and III. Pin "H" of the generator connector shall be capable of supplying 0.5 A at a voltage greater than or equal to (E_r) minus 0.25 V over the operating temperature range of the regulator. (see 4.5.14). For type III only, this requirement shall be met with pin "E" of the battery connector grounded and ungrounded.

3.4.16 Electromagnetic interference (EMI). The regulator shall limit emissions for tactical vehicle components as specified in 3.4.16.1 and 3.4.16.2 (see 4.5.15).

3.4.16.1 Radiated emissions. The radiated emissions shall meet the requirements of figure 6 (see 4.5.15).

3.4.16.2 Conducted emissions. On a 28 Vdc, 50 A lead of the split load, the conducted emissions shall meet the requirements of figure 7 (see 4.5.15).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:50Z
Check the source to verify that this is the current version before use.

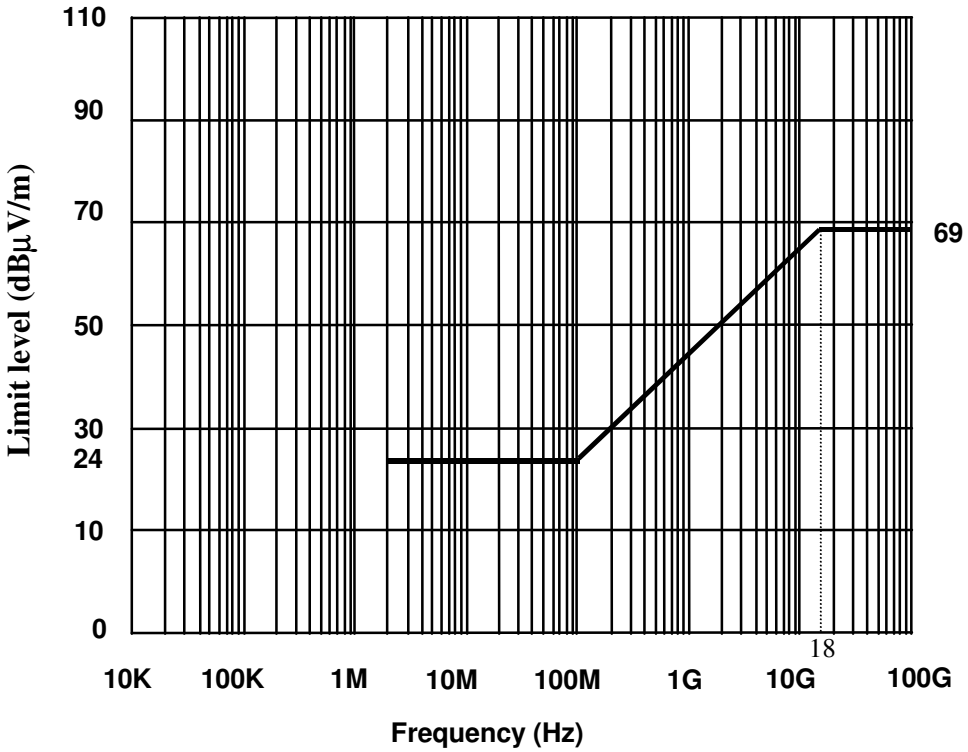


FIGURE 6. Radiated emission.

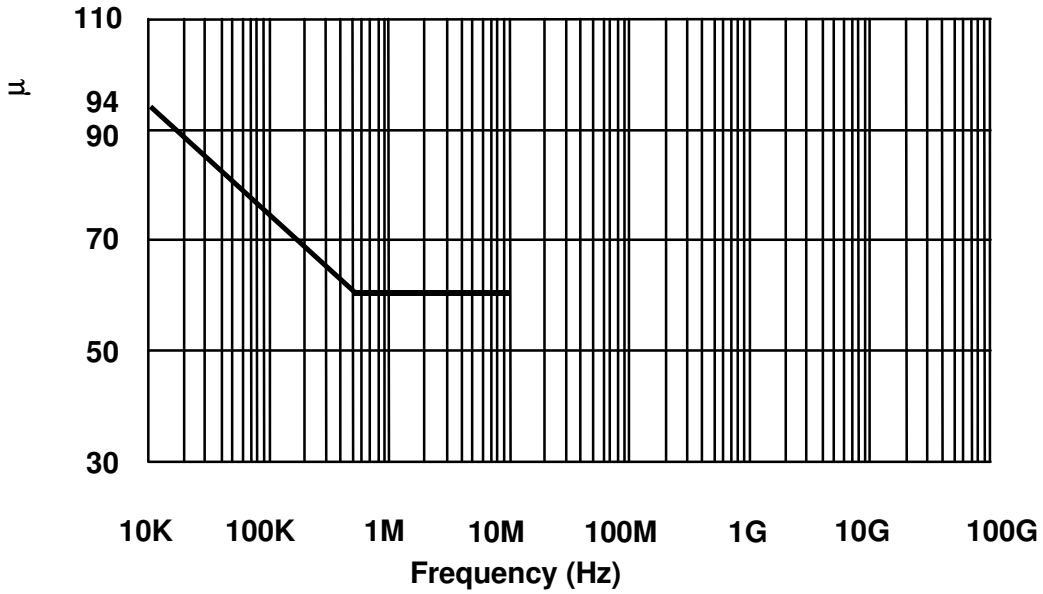


FIGURE 7. Conducted emission.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:50Z
Check the source to verify that this is the current version before use.

3.5 Interface requirements.

3.5.1 Weight.

3.5.1.1 Type I. Not applicable.

3.5.1.2 Type II. The regulator shall not be more than 19 pounds (lb) (8.6 kilograms (kg)) (see 4.6.1).

3.5.1.3 Type III. The regulator shall not be more than 19 lb (8.6 kg) (see 4.6.1).

3.6 Support and ownership requirements.

3.6.1 Reliability. Quantitative reliability numbers are to be determined by Government reliability engineers prior to finalizing the contract.

3.6.2 Endurance.

3.6.2.1 Type I. The regulator shall withstand 2000 hours of operation under the load, speed and temperature conditions specified herein without servicing, adjustment or replacement parts (see 4.7.1.1).

3.6.2.2 Types II and III. The regulator shall withstand 1000 hours of operation under the load, speed and temperature conditions specified herein without servicing, adjustment or replacement parts (see 4.7.1.2).

3.6.3 Test equipment. Existing Government or commercial test equipment used in conjunction with the test item shall be identified along with the requirement for compatibility between the item and the test equipment.

3.6.4 Transportation. The appropriate packaging shall be applied for domestic and/or overseas destinations (see 5.1).

3.6.5 Identification and marking.

3.6.5.1 Marking. Identification markings shall conform with ANSI/SAE AS478.

3.6.5.2 Nameplate. A nameplate shall be attached to the regulator housing in the manner specified on the applicable standard or drawing. The legend contained on the plate shall include the following (see 4.4):

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:50Z
Check the source to verify that this is the current version before use.

a. Regulator
b. 28 Vdc
c. Manufacturer's identification
d. Serial number
e. Military part number
f. Federal stock number
g. Manufacturer's part number
h. Date of manufacture
i. Contract number
j. US

3.6.6 Workmanship. Workmanship shall be of a quality which assures a product free of burrs, rust, scratches, chips, sharp edges, loose or defective connectors, cracked insulation, faulty soldering, or other defects which adversely affect serviceability or appearance (see 4.4.4).

3.7 Environmental requirements.

3.7.1 High temperature.

3.7.1.1 Type I. The regulator shall operate at a temperature of 225° ± 5°F (107° ± 3°C), except that the output voltage shall be 26.4 ± 0.8 V and the output current shall be 22 ± 2 A (see 4.8.1.1).

3.7.1.2 Types II and III. The regulator shall operate at a temperature of 215° ± 5°F (101° ± 3°C), except that the output voltage shall be 26.5 ± 0.7 V (see 4.8.1.2).

3.7.2 Low temperature.

3.7.2.1 Type I. The regulator shall operate at a temperature of -65° ± 5°F (-54° ± 3°C) and the output voltage shall be 29.6 ± 0.8 V and the output current shall be 35 ± 3 A (see 4.8.2.1).

3.7.2.2 Types II and III. The regulator shall operate at a temperature of -65° ± 5°F, and the output voltage shall be 29.5 ± 0.7 V (see 4.8.2.2).

3.7.3 Vibration. The regulator shall not be damaged nor performance degraded after exposure to vibration levels in accordance with MIL-STD-202, method 204D, condition A, or equivalent (see 4.4.1), encountered during regulator operation (see 4.8.3).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:50Z
Check the source to verify that this is the current version before use.

3.7.4 Shock. The regulator shall not be damaged nor performance degraded after exposure to shock levels in accordance with MIL-STD-202, method 213B, condition A, or equivalent (see 4.4.1), encountered during regulator operation (see 4.8.4).

3.7.5 Corrosion resistance. The regulator shall evidence no corrosion that would adversely affect performance after being subjected to a salt spray corrosion test specified in ASTM B117 for 200 hours (see 4.8.5).

3.7.6 Fungus resistance. The regulator shall not support microbial growth that would adversely affect performance after being exposed to fungi for 90 days (see 4.8.6).

3.7.7 Waterproofness. The regulator shall show no evidence of leakage nor performance degradation after being subjected to the tests specified in 4.8.7.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspections (see 4.3).

4.2 First article inspection. Unless otherwise specified (see 6.2), the Government shall select four regulator assemblies produced under the production contract for first article inspection. The selected regulators shall be subjected to the first article tests specified in table I and in the order listed in table II. Approval of the first article sample by the Government shall not relieve the contractor of his obligation to supply assemblies that are fully representative of those inspected as a first article sample. Any changes or deviation of the production units from the first article sample shall be subject to the approval of the contracting officer (see 3.1).

4.3 Conformance inspections.

4.3.1 Sampling. Sampling requirements, if any, shall be specified in the contract (see 6.2).

4.3.2 Examination. For examination purposes, defects shall be classified as listed in table III.

4.3.3 Tests. Samples selected in accordance with 4.3.1 shall be subjected to the conformance tests as specified in table I, and in the order listed in tables IV and V.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:50Z
Check the source to verify that this is the current version before use.

TABLE I. Classification of inspections.

Title	Requirement	Inspection	First article		Conformance		
					Exami-	Tests	
			1/	2/	nation	1/	2/
Materials and construction	3.2, 3.3 thru 3.3.6	4.4.3, 4.4	X		X		
Operating requirements	3.4	4.5					
Polarity	3.4.1	4.4	X		X		
Reverse current	3.4.2	4.5.1	X			X	
Transient voltage	3.4.3	4.5.2	X	X			
Load switching	3.4.4	4.5.3	X	X			
Voltage regulation E_R	3.4.5	4.5.4	X	X		X	X
Voltage ripple	3.4.6	4.5.5	X				
Current regulation	3.4.7	4.5.6	X	X		X	X
Overvoltage	3.4.8	4.5.7	X				X
Field excitation	3.4.9	4.5.8	X				X
Effective field circuit resistance	3.4.10	4.5.9	X	X			
Negative voltage operation	3.4.11	4.5.10	X	X			X
Field circuit	3.4.12	4.5.11	X	X			
Reverse battery polarity	3.4.13	4.5.12	X	X			
Short circuit protection	3.4.14	4.5.13		X			X
Generator accessory sensing circuit	3.4.15	4.5.14		X			X
Electromagnetic interference (EMI)	3.4.16	4.5.15	X	X			
Interface requirements	3.5	4.6					
Weight	3.5.1	4.6.1	X				
Support and ownership requirements	3.6	4.7					
Endurance	3.6.2	4.7.1	X	X			
Environment requirements	3.7	4.8					
High temperature	3.7.1	4.8.1	X	X			
Low temperature	3.7.2	4.8.2	X	X			
Vibration	3.7.3	4.8.3	X	X			
Shock	3.7.4	4.8.4	X	X			
Corrosion resistance	3.7.5	4.8.5	X	X			
Fungus resistance	3.7.6	4.8.6	X	X			
Waterproofness	3.7.7	4.8.7	X	X		X	X

1/ Type I regulator.
2/ Types II and III regulator.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:50Z
Check the source to verify that this is the current version before use.

TABLE II. Sequence of first article inspection.

Test	Paragraph	Sample			
		1	2	3	4
Reverse current	4.5.1	X	X	X	X
Voltage regulation	4.5.4	X	X	X	X
Voltage ripple	4.5.5	X	X	X	
Current regulation	4.5.6	X	X	X	
Field excitation	4.5.8	X	X	X	
High temperature	4.8.1		X	X	
Low temperature	4.8.2		X	X	
Endurance	4.7.1	X			
Vibration	4.8.3		X	X	
Shock	4.8.4		X	X	
Waterproofness	4.8.7	X	X	X	
Corrosion resistance	4.8.5		X		
Transient voltage	4.5.2		X	X	
Overvoltage	4.5.7	X	X	X	
Fungus resistance	4.8.6			X	
Electromagnetic interference	4.5.15				X
Field circuit	4.5.11	X	X	X	
Negative voltage operation	4.5.10	X	X		
Short circuit protection	4.5.13	X	X	X	
Generator accessory sensing circuit	4.5.14	X	X	X	

4.4 Verification methods. Acceptable verification methods included in this section are visual inspection and measurement, sample tests, full-scale demonstration tests, simulation, modeling, engineering evaluation, component properties analysis, and similarity to previously-approved or previously-qualified designs.

4.4.1 Verification alternatives. The manufacturer may propose alternative test methods, techniques, or equipment, including the application of statistical process control, tool control, or cost effective sampling procedures to verify performance. See the contract for alternatives that replace verification methods required by this specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:50Z
Check the source to verify that this is the current version before use.

4.4.2 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be conducted under the following conditions:

a. Air temperature: 73 ± 18°F (23 ± 10°C)
b. Barometric pressure: 28.5 (+2, -3) in. of mercury (Hg)
   96.5 (+6.77, -10) kilopascals (kPa)
c. Relative humidity: 50 ± 30%

TABLE III. Classification of defects.

Category	Defect	Method of examination
Critical	None	
Major:		
101	Assembly incomplete (see 3.3).	Visual
102	Nonconformance in design and construction (see 3.3.6 and 3.4.1).	Visual
103	Dimensions affecting interchangeability, out of tolerance (see 3.3.1, 3.3.6).	Measurement
104	Finish, improper application (see 3.2.2).	Measurement
105	Identification marking improper (see 3.6.5).	Visual
106	Faulty workmanship affecting performance (see 3.2).	Visual
Minor:		
201	Dimensions not affecting interchangeability, out of tolerance (see 3.3.1).	Visual
202	Faulty workmanship affecting appearance (see 3.6.6).	Visual

TABLE IV. Sequence of conformance tests (type I).

Test	Paragraph
Reverse current	4.5.1
Voltage regulation	4.5.4
Current regulation	4.5.6
Waterproofness	4.8.7

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:50Z
Check the source to verify that this is the current version before use.

TABLE V. Sequence of conformance tests (types II and III).

Test	Paragraph
Voltage regulation	4.5.4
Current regulation	4.5.6
Generator accessory sensing circuit	4.5.14
Overvoltage	4.5.7
Field excitation	4.5.8
Negative voltage operation	4.5.10
Waterproofness	4.8.7
Short circuit protection	4.5.13

4.4.2.1 Type I. Unless otherwise specified herein, the test circuit shall include the regulator, battery, and generator. The generator employed in the test circuit shall conform to Drawing 7355736 or Drawing 10950808. The battery shall consist of two 12 V batteries conforming to type 2 HN of MIL-B-11188 or equivalent. The two batteries shall be connected in series. Batteries shall be fully charged (see 6.5.2.1) and shall be kept at room temperature.

4.4.2.2 Types II and III. The test circuit as shown in figure 5 shall include the regulator, mounted on a horizontal plane, battery and generator. The generator used in the test circuit shall conform to MIL-PRF-62061. The batteries shall consist of six 12 V batteries conforming to type 6TL of MIL-B-62346 or equivalent. The six batteries shall be connected in series-parallel. Batteries shall be fully charged (see 6.5.2.2), and kept at room temperature.

4.4.3 Materials and construction. The materials and construction shall be determined by inspection of contractor records providing proof or certification that design, construction, processing, and materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.5 Operating requirements verification.

4.5.1 Reverse current. The stabilized regulator (see 6.5.1) shall be connected to the test circuit with the generator operating at sufficient speed to deliver charging current to the battery. The generator speed shall then be gradually reduced to zero. The maximum value of the reverse current shall be observed on the ammeter or milliammeter. Five test cycles shall be observed.

4.5.2 Transient voltage.

4.5.2.1 Type I. The regulator shall be connected in the test circuit with the battery disconnected and the generating system inoperative. A negative voltage pulse of 100 V, square wave, of 50 ms duration, shall be applied to the regulator output terminal, once every 2 seconds,

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:50Z
Check the source to verify that this is the current version before use.

equally spaced, for a period of 10 minutes. The test shall then be repeated, except that the voltage pulse shall be positive and the period shall be 5 minutes. At the conclusion of this test, the regulator shall pass the tests of 4.5.1, 4.5.4.1, 4.5.5, and 4.5.6.1.

4.5.2.2 Types II and III. The regulator shall be disconnected from the test circuit as shown in figure 5 and stored for 24 hours at 215° ± 5°F. It shall then be tested in an ambient temperature of 215° ± 5°F as follows. A voltage pulse of 100 V, square wave, not more than 160 microseconds rise time, of 50 ms duration shall be applied to the regulator output terminal once every 6 seconds, equally spaced for a period of 3 hours. The test shall be repeated except that a voltage pulse of 70 V shall be applied to the input terminal of the regulator.

4.5.2.2.1 Negative transient voltage. The regulator shall be temperature stabilized at 215°± 5°F for a period of one hour before being subjected to five cycles of the following two-step test procedure, using the test circuit shown in figure 8. Steps one and two shall constitute a cycle and the cycle shall be repeated five times.

Step 1:
a. Capacitor C shall be charged to a voltage (E_C) of 595 ± 5 V.
b. Switch S_2 shall be closed (E_C is applied to the input terminal of the regulator) for a period of 6 seconds and then opened.

Step 2:
a. With the generator running at a speed between 2400 and 3000 rpm, switch S_3 shall be closed for 1 minute minimum.
b. Capacitor C shall be charged to an E_C of 595 ± 5 V and switch S_2 closed for a period of 6 seconds minimum.
c. Open all switches.

4.5.3 Load switching.

4.5.3.1 Type I. The regulator shall be connected in the test circuit with the battery disconnected. With the generator speed at 5000 rpm, the resistance load shall be adjusted to deliver a current regulated load of 28 V at -65°F, 27 V at 77°F and 25 V at +225°F. The load shall be switched on for one second, off for one second, and repeated for a period of 15 minutes. Load switching time shall be 1.5 ± 0.5 ms. At the conclusion of this test, the regulator shall pass the test of 4.5.1, 4.5.4.1 and 4.5.6.1.

4.5.3.2 Types II and III. Not applicable.

4.5.4 Voltage regulation.

4.5.4.1 Type I. The operationally stabilized regulator (see 6.5.1.1) shall be connected in the test circuit. During the test, the resistance load shall be applied and removed suddenly. Observations shall be made of the output voltage while the regulator is being subjected to the following test procedure:

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:50Z
Check the source to verify that this is the current version before use.

a. Increase generator speed from zero, and operate generator at 2000 rpm with resistance load at zero.
b. With generator at 2000 rpm, add resistance load to establish a total load of 20 amps.
c. Increase generator speed to 5000 rpm and maintain the 20 A load.
d. With generator at 5000 rpm, remove the resistance load.
e. Increase generator speed to 8000 rpm with resistance load at zero.
f. With generator at 8000 rpm, add resistance load to establish a total load of 20 A.
g. Decrease generator speed to 2000 rpm and maintain the 20 A load.
h. With generator at 2000 rpm, remove the resistance load.
i. Increase generator speed to 8000 rpm with resistance load at zero.

4.5.4.2 Types II and III (first article). The temperature stabilized regulator (see 6.5.1.2) shall be connected in the test circuit. During the test, the resistance load shall be applied and removed suddenly. The change in load shall be accomplished in 10 ms maximum time. Tripping of overvoltage reset due to quick reaction shall constitute failure. Observation shall be made of the output voltage while the regulator is being subjected to the following test procedure.

	RPM	Load(A)
a.	2400 ± 50	25
	2400 ± 50	100
	2400 ± 50	350
b.	5500 ± 110	same sequence as in (a)
c.	8000 ± 160	same sequence as in (a)

E_R to be measured on any recording device having a bandwidth of 5 kilohertz or greater in order to obtain the time and amplitude as shown in figure 1.

4.5.4.3 Types II and III (conformance). The generator shall be operated at 5500 rpm with a 350 A load. The regulated voltage shall be determined with the regulator at an ambient temperature of 80 ± 15°F under the following conditions.

RPM	Load (A)
2400 ± 50	Battery load only
2400 ± 50	350
8000 ± 160	Battery load only
8000 ± 160	350

4.5.5 Voltage ripple (types II and III). Using the test procedure specified in 4.5.4.2 and 4.5.4.3, the output voltage waveform at the regulator output shall be measured with an oscilloscope as shown in figure 5. Repeat the procedure with batteries disconnected.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:50Z
Check the source to verify that this is the current version before use.

4.5.6 Current regulation.

4.5.6.1 Type I. With the regulator operationally stabilized (see 6.5.1.1), the generator operating at 5000 rpm, and the battery disconnected from the test circuit, the resistance load shall be adjusted to establish current regulation. The resistance load shall be further adjusted until the voltage output has been reduced from a maximum voltage 28 V at -65°F (-54°C), 27 V at 77°F (25°C), and 25 V at 225°F (107°C) to 6 V. An ammeter shall be observed to verify current reading. Below 18 V, only the maximum current limit value shall apply (38 A at -65°F, 30 A at 77°F, and 24 A at 225°F). The procedure shall be repeated except that the battery shall be connected in the test circuit and the ammeter shall be observed for current reading at 25 V.

4.5.6.2 Types II (first article) and III (pin "E" ungrounded). The regulator shall be operated, without batteries connected, at a generator speed at 2400 rpm. Resistive loads of 0.08, 0.07, 0.06, 0.05, 0.04, 0.03, and 0.028 Ω shall be applied and generator current (I_G) shall be observed for each of the seven loads. Then a 0.028 Ω resistive load shall be suddenly applied and released. Load application time and load release time shall not exceed 10 ms. Repeat this test at 5500 rpm and 8000 rpm.

4.5.6.3 Types II (conformance) and III (pin "E" ungrounded). The regulator shall be operated, without batteries connected, with the generator speed at 5500 rpm. I_G shall be observed with a 0.08 Ω load applied and again with a 0.028 Ω applied.

4.5.6.4 Type III (pin "E" grounded) (first article). The batteries shall be operated, with pin "E" of the battery connector grounded, without batteries connected, at a generator speed of 3000 rpm. Resistive loads of 0.28, 0.20, 0.17, 0.14, 0.11 and 0.09 Ω shall be applied and generator current (Ig) shall be observed for each of the six loads. Then a 0.09 Ω resistive load shall be suddenly applied and released. Load application time and load release time shall not exceed 10 ms.

4.5.6.5 Type III (pin "E" grounded) (conformance). The regulator shall be operated, with pin "E" of the battery connector grounded, without batteries connected, with the generator speed at 3000 rpm. Ig shall be observed with a 0.28 Ω load applied and again with a 0.09 Ω applied.

4.5.7 Overvoltage.

4.5.7.1 First article. The regulator shall be connected to a dc power supply. The voltage shall be increased from 28 V by application of a step voltage of approximately 6 V. The voltage level and reaction time shall be measured. Failure of the regulator to deactivate the generator when the system voltage reaches 33 $\pm$ 1 V within a time of no less than 0.25 second and no more than 1 second shall be cause for rejection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:50Z
Check the source to verify that this is the current version before use.

4.5.7.2 Conformance. The regulator shall be connected to a dc power supply. With 32 V applied, the overvoltage protection circuit shall not activate. With 34 V applied, the overvoltage protection circuit shall activate.

4.5.8 Field excitation. The regulator shall be connected in the circuit of figure 5 with batteries disconnected, S1 open and the generator operating at constant speed. An external power supply shall be momentarily connected to the generator field with its polarity reversed. This momentary connection shall be repeated until the residual voltage is less than 0.7 V. The residual voltage shall be measured. The regulator shall excite and bootstrap to its nominal output when S1 is closed.

4.5.9 Effective field circuit resistance (type I). The operationally stabilized regulator shall be connected in the test circuit with battery disconnected. The generator speed shall be gradually increased from zero to produce a field current of 0.8 A in the unregulated mode and a load current adjusted to 18 A. The field current and generator-to-field voltage drop shall be measured and the effective field circuit resistance shall be determined.

4.5.10 Negative voltage.

4.5.10.1 Negative voltage operation (types II and III) (first article). The regulator shall be operated until temperature stabilization. Then it shall be subjected to the following test procedure using the test circuit shown in figure 9 (batteries shall be fully charged).

a. Arrange the test set-up as shown on the circuit diagram on figure 9 with S1, S2, and S3 open.
b. Close S2 for 1 to 5 seconds, then open S2.
c. Close S3 with generator at 2400 $\pm$ 50 rpm. Observe generator output (I_G), which should be less than 200 A. If I_G indicates 200 A or more, open S3 and discontinue test due to failure. If I_G does not exceed 200 A, close S1 and test regulator in accordance with 4.5.4.2, 4.5.6.2, and 4.5.7.
d. Open S1, S2, and S3 . Lock generator to prevent rotation (rotation less than 10 rpm is permissible). Close S3 and reverse the polarity of V2 for 10 seconds, then return polarity to normal. Repeat step (c) and determine that output is as specified. Momentary excursion of I_g beyond 200 A is permissible.

NOTE: Steps (b) and (c) test for ability to withstand reverse generator operation. Step (d) tests the ability to withstand reverse slave battery operation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:50Z
Check the source to verify that this is the current version before use.

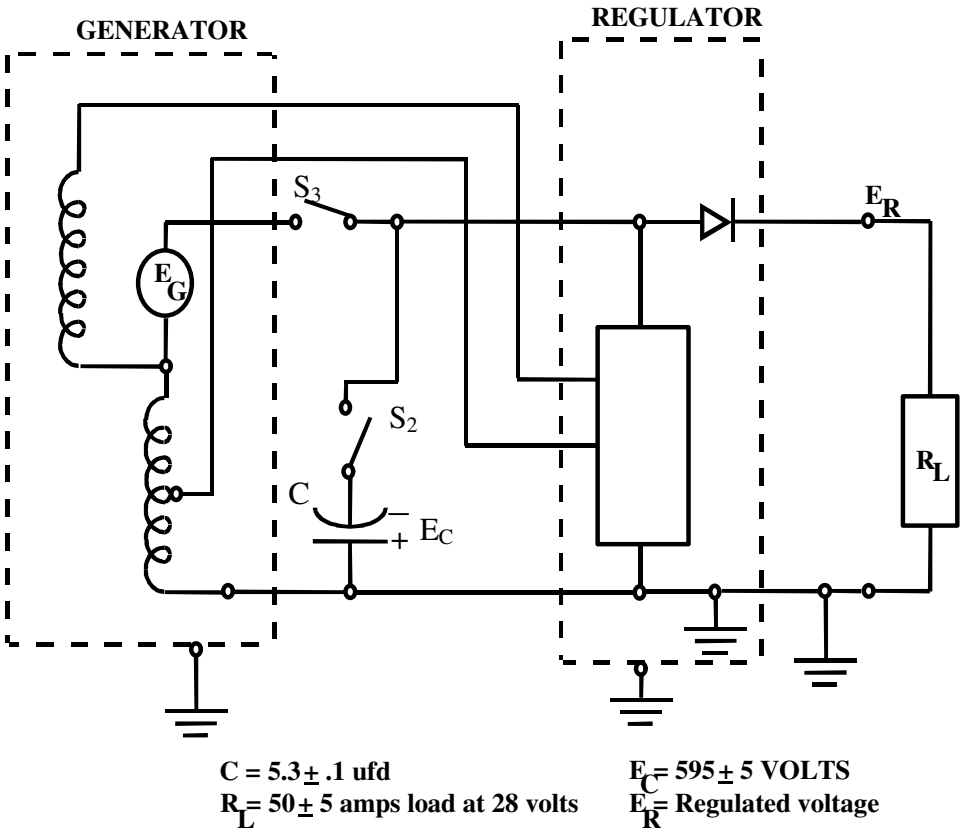


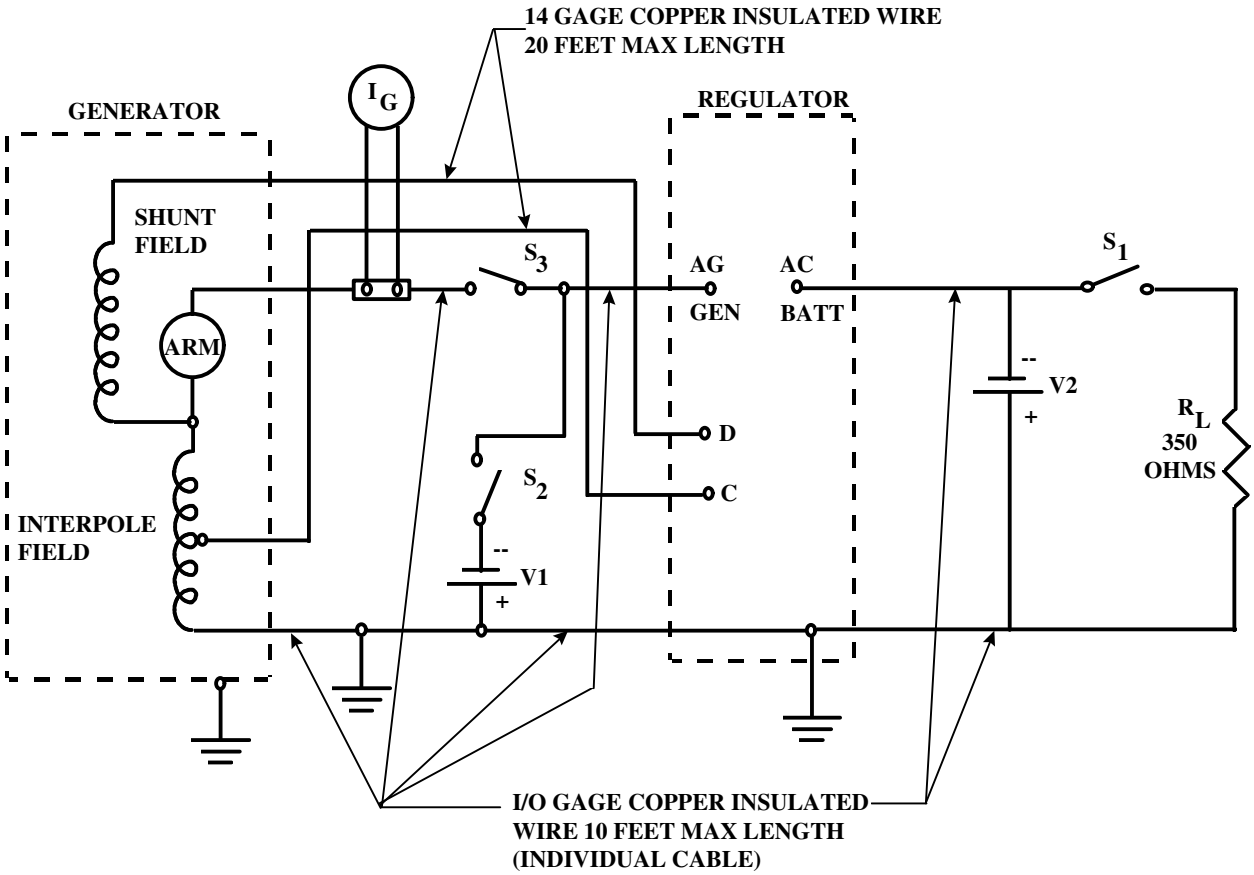
Figure 8. Test circuit.

4.5.10.2 Negative voltage operation (conformance). The regulator shall be subjected to the following test procedure using the test circuit of figure 9:

a. Close S2 from 1 to 5 seconds, then open.
b. Lock the generator to prevent rotation (rotation of less than 10 rpm is permissible). Close S3 and reverse the polarity of V2 for 10 seconds. Return polarity to normal.

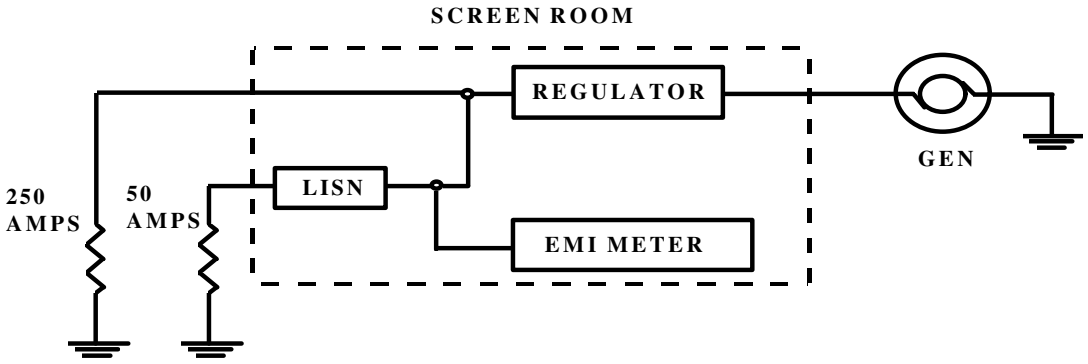
NOTE: Step (a) tests for ability to withstand reverse generator operation. Step (b) tests for ability to withstand reverse slave battery operation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:50Z
Check the source to verify that this is the current version before use.



$V_1 = V_2 = 24$ VDC (TWO 12V - 6TN BATTERIES, OR EQUIVALENT)

FIGURE 9. Negative voltage circuit.



LISN - LINE IMPEDANCE STABILIZATION NETWORK

FIGURE 10. Electromagnetic interference tests

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:50Z
Check the source to verify that this is the current version before use.

4.5.11 Field circuit (types II and III). The regulator shall be tested as follows:

a. With the generator operating at 2400 rpm in the circuit shown on figure 5 with a 30 A load, switch S1 shall be opened and closed 5 times. The load shall be increased to 400 A and the procedure repeated.
b. Procedure (a) shall be repeated with the generator operating at 8000 rpm.

4.5.12 Reverse battery polarity. The regulator shall be connected in the test circuit. With the generator inoperative, the battery polarity shall be reversed and a voltage applied in increments from maximum voltage to a minimum of 6 V. During the tests, checks for open circuit shall be made.

4.5.13 Short circuit protection. With the regulator connected and operating in the test circuit shown on figure 5; pin "D" of the generator connector shall be connected to ground and this connection shall be maintained for ten (10) seconds minimum. The ground connection shall be removed, the reset switch shall be depressed, and this test shall be repeated for pin "H" of the generator connector. For type III only, this test shall be repeated for pin "F" of the generator connector. Subsequent to this test the regulator shall meet the requirements of 4.5.4.

4.5.14 Generator accessory sensing circuit. A resistive load shall be connected to pin "H" and the regulator shall be operated in the test circuit of figure 5. With a minimum of 0.5 A load current on pin "H", the voltage on pin "H" shall be observed and shall be within the specified limits.

4.5.15 Electromagnetic interference. The regulator shall be tested in accordance with MIL-STD-462 or equivalent (see 4.4.1) and figure 10.

4.6 Interface requirements verification

4.6.1 Weight (types II and III). Use a scale to verify that the weight of the regulator shall not exceed 19 lbs.

4.7 Support and ownership requirements verification.

4.7.1 Endurance.

4.7.1.1 Type I. The regulator shall function continuously when cycled in accordance with figure 11 at 24 cycles per hour for 2000 hours. Each 1000 hour test period shall be divided into time-temperature periods; and each time-temperature period shall be divided between different resistance loads as specified in table VI. After each 1000 hour test period, the regulator shall pass the test of 4.5.1, 4.5.3.1, and 4.5.4 through 4.5.6.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:50Z
Check the source to verify that this is the current version before use.

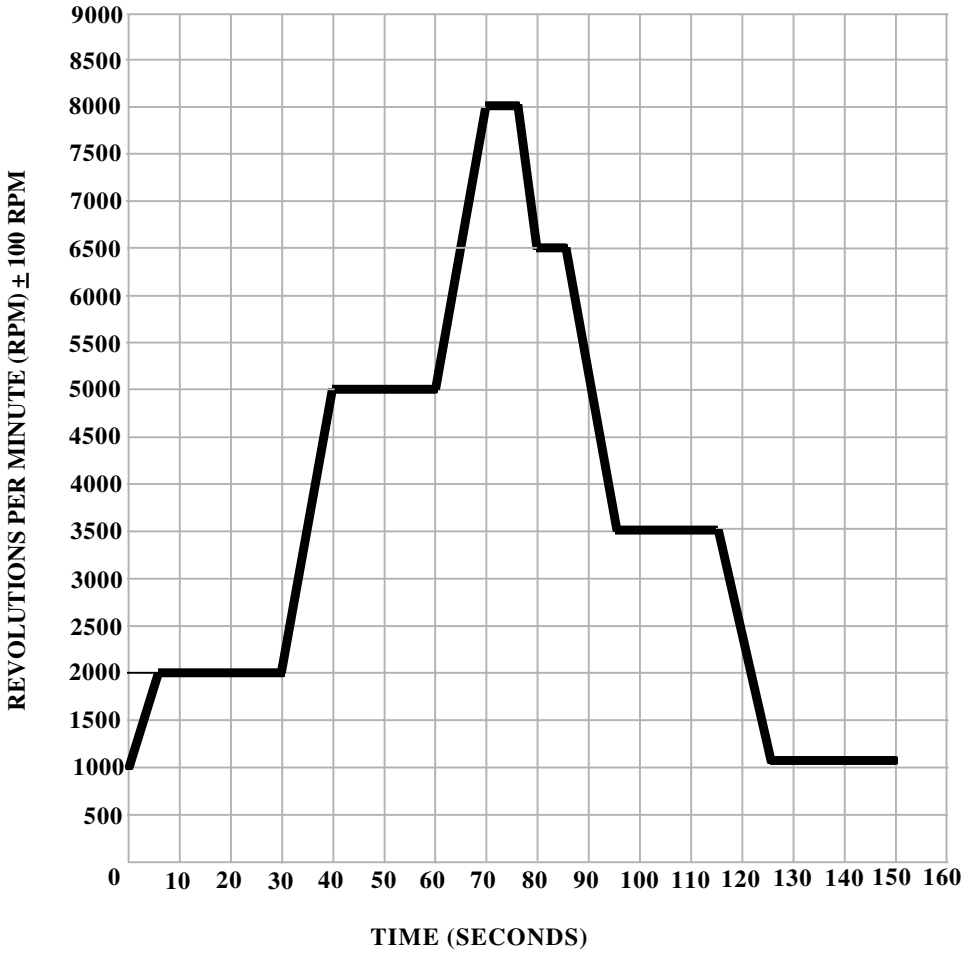


FIGURE 11. Endurance cycle.

TABLE VI. Endurance test

Time (hours)	Temperature (°F)	Resistance load	
		One-half of time temperature period (amperes)	One-half of time temperature period (amperes)
190	225	18	4
190	77	23.5	4
50	-65	29	4
140	0	27	4
190	77	23.5	4
240	160	20	4

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:50Z
Check the source to verify that this is the current version before use.

4.7.1.2 Types II and III. The regulator shall be cycled at 18 cycles per hour for 1000 hours. The 1000 hours shall be divided into 500 hours with generator operating at 2400 rpm. 300 hours at 4000 rpm, 150 hours at 6000 rpm and 50 hours at 8000 rpm. A cycle shall consist of generator operation for 25 seconds each at battery load, 100 A, 200 A, 300A, 400 A, followed by a reduction to 300 A, 200 A, 100 A and then at battery load. At the conclusion of this test, the regulator shall pass the tests of 4.5.1 and 4.5.4 through 4.5.6.

4.8 Environmental requirements verification

4.8.1 High temperature.

4.8.1.1 Type I. The regulator shall be stored for 24 hours in an ambient temperature of 225 ± 5°F. With ambient temperature maintained at 225 ± 5°F, the regulator shall be tested to meet the requirements of 3.4.1, 3.4.3.1, 3.4.4.1, and 3.4.9.1. While remaining at the same ambient temperature and connected in the test circuit, the regulator shall function continuously with a field current of 0.75 A for one-half hour at each of the following conditions:

a. Output voltage of 25 V in the current regulated mode.
b. Load current of 12.5 A in the voltage regulated mode.

Then the regulator shall be returned to an ambient room temperature of 77° ± 15°F and shall pass the tests specified in 4.5.1, 4.5.4.1, and 4.5.6.1.

4.8.1.2 Types II and III. The regulator shall be stored for 24 hours in an ambient temperature of 215 ± 5°F. With ambient temperature maintained at 215 ± 5°F, the regulator shall function continuously for 4 hours while the generator speed and output load are adjusted to produce 2400 rpm and 400 A. While remaining at the same ambient temperature, the regulator shall be tested as specified in 4.5.1, 4.5.4.1, 4.5.7, 4.5.8, 4.5.10 and 4.5.2.2. The regulator shall be returned to an ambient temperature of 80° ± 5°F and shall pass the tests specified in 4.5.1, 4.5.6.2, 4.5.7, and 4.5.11.

4.8.2 Low temperature.

4.8.2.1 Type I. The regulator shall be stored for 24 hours in an ambient temperature of -80° ± 5°F. The regulator shall then be tested in the following order in an ambient temperature of -65° ± 5°F in accordance with 4.5.1, 4.5.2.1, 4.5.3.1, 4.5.4, 4.5.5, 4.5.8, 4.5.9, and 4.5.12.

4.8.2.2 Types II and III. The regulator shall be stored for 24 hours in an ambient temperature of -80° ± 5°F. After stabilizing for 4 hours at -65° ± 5°F, the regulator shall function continuously for 1 hour while the generator speed and output load are adjusted to produce 2400 rpm and 400 A. While remaining in an ambient temperature of -65° ± 5°F, the regulator

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:50Z
Check the source to verify that this is the current version before use.

shall be tested as specified in 4.5.1, 4.5.2, 4.5.4, 4.5.7, 4.5.8, and 4.5.10. The regulator shall be returned to an ambient temperature of +80° ± 5°F and shall pass the tests specified in 4.5.1, 4.5.6.2, and 4.5.7.

4.8.3 Vibration. The regulator shall be tested in accordance with MIL-STD-202, method 204D, condition A, or equivalent (see 4.4.1) except that the sweep time shall be performed 24 times. At the conclusion of this test, the regulator shall pass the tests of 4.5.1, 4.5.4, 4.5.5, and 4.5.6.

4.8.4 Shock. The regulator shall be tested in accordance with MIL-STD-202, method 213B, condition A, or equivalent (see 4.4.1). At the conclusion of this test, the regulator shall pass the tests of 4.5.1, 4.5.4, 4.5.5, and 4.5.6.

4.8.5 Corrosion protection. A test sample shall be prepared consisting of hot dipped galvanized steel in accordance with ASTM A123, with a minimum zinc coating thickness of G45 to G55 (nominal 2 mils) on sheet steel less than 0.06 in. thick, 2.5 mils on sheet steel greater than 0.06 in. thick or 0.75 mil on pre-galvanized sheet steel 0.06 in. thick or less in accordance with ASTM A924, which refers to ASTM A653 thickness G60. A representative regulator and the zinc galvanized sample shall be subjected to the accelerated corrosion test (ACT) in accordance with ASTM B117 for 200 hours exposure. Following the corrosion test, the regulator and the zinc galvanized sample shall be examined for corrosion. The corrosion tested regulator shall meet or exceed the corrosion resistance provided by the zinc galvanized sample and shall pass the tests of 4.5.1, 4.5.4, 4.5.5, and 4.5.6.

4.8.6 Fungus resistance. The regulator shall be tested in accordance with ASTM G21 except that the period of exposure shall be a continuous 90 days. At the conclusion of this test, the regulator shall pass the tests of 4.5.1, 4.5.4, 4.5.5, and 4.5.6.

4.8.7 Waterproofness. The regulator shall be tested as specified herein.

4.8.7.1 Type I (nonoperational). The regulator shall be submerged in a container with the uppermost surface a minimum of 1 in. below the surface of the tap water solution and installed in the chamber. Electrical operation during submersion is not required. The chamber shall be evacuated to a pressure 6 psi (41 kPa) below atmospheric so as to apply a minimum of 6 psi in internal pressure to all voids within the regulator. Any evidence of bubbles escaping from the interior of the regulator shall constitute a failure of this test. Bubbles which are the result of trapped air on the exterior surfaces of the regulator shall not be cause for failure. The chamber shall then be pressurized to 6 psi above atmospheric and the regulator again operated for 30 minutes. At the conclusion of this test, the regulator shall pass the tests of 4.5.1, 4.5.4, 4.5.5 and 4.5.6.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:50Z
Check the source to verify that this is the current version before use.

4.8.7.2 Types II and III (operational) first article. The regulator shall be tested as specified in 4.8.7.1, except salt water solution shall be used instead of tap water, 10 psi (68 kPa) pressure and 10 psi (68 kPa) vacuum instead of 6 psi (41 kPa) shall be applied. During submersion, the regulator shall be subjected to 10 psig for a period of 5 minutes. The regulator while submerged shall be operable and shall pass the test in accordance with 4.5.4.2 during the pressure and vacuum tests phases.

4.8.7.3 Types II and III (nonoperational) conformance. For conformance tests, non-operating regulators shall be tested as specified in 4.8.7.2, except 10 psi (68 kPa) pressure shall be applied for a period of 1 minute instead of 5 munites.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. The regulators covered by this specification are military unique because they are intended for use in 28 V dc nominal electrical systems equipped with charging generators for non-commercial tactical and combat military heavy-duty vehicles. Because of the strict criteria and physical characteristics, the regulator has no commercial application.

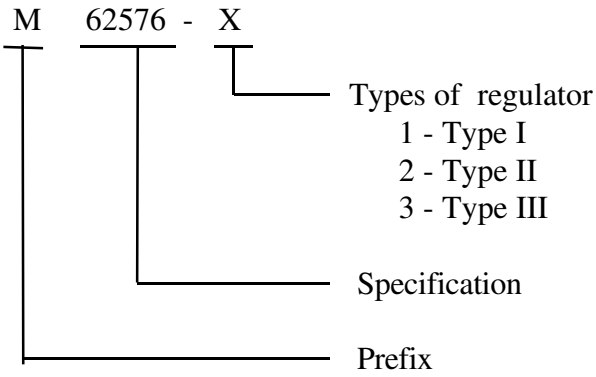
6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1 and 2.3).
c. When first article is required (see 3.1).
d. Title, number, and date of the applicable drawing (see 3.3.1).
e. If sample size for first article inspection should be other than as specified (see 4.2).
f. If sampling is required (see 4.3.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:50Z
Check the source to verify that this is the current version before use.

g. If inspection conditions should be other than as specified (see 4.4.2).
h. Packaging requirements (see 5.1).

6.3 Part or Identifying Number (PIN). The PIN to be used for regulators acquired to this specification are created as follows:



6.4 Subject term (key word) listing.

DC solid state
Dual current limit
Single current limit
Steady state voltage
Voltage controller

6.5 Definitions.

6.5.1 Stabilized regulator.

6.5.1.1 Type I. A temperature stabilized regulator is defined as a regulator which has operated for one hour with the generator speed adjusted to 3000 rpm and the load adjusted to 12.5 A (see 4.5.4.1).

6.5.1.2 Types II and III. A temperature stabilized regulator is defined as a regulator which has been operated for 30 minutes with the generator speed adjusted to 2400 rpm, the load adjusted to 175 A and at ambient room temperature (see 4.5.4.2).

6.5.2 Fully charged battery.

6.5.2.1 Type I. For the purposes of the test procedures in this specification, a fully charged battery is defined as a battery having a current consumption of not more than 2 A at 28 V at ambient room temperature (see 4.4.2.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:50Z
Check the source to verify that this is the current version before use.

6.5.2.2 Types II and III. For the purposes of the test procedures in this specification, a fully charged battery is defined as a battery having a current consumption of not more than 4 A at 28 V at ambient room temperature (see 4.4.2.2).

6.5.3 Load current. Load current, for purposes of this specification, should be considered as that current from generator and regulator to batteries and resistive load (see figure 5).

6.6 Cross-reference of classification. Type I regulator replaces the cancelled regulator specification MIL-R-62067(AT), type II regulator replaces the cancelled regulator specification MIL-R-62104(AT), and type III regulator replaces the cancelled ATPD 2062.

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - AT
Navy - YD1

Review Activities:
Army - MI
DLA - CC

Preparing Activity:
Army - AT

(Project 2920-0491)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:50Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-DTL-62576B	**2. DOCUMENT DATE *(YYMMDD)*** 980930

3. DOCUMENT TITLE Regulator, Engine Generator

4. NATURE OF CHANGE ***(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)***

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	**b. ORGANIZATION**	
c. ADDRESS *(Include Zip Code)*	**d. TELEPHONE *(Include Area Code)*** **(1) Commercial** **(2) AUTOVON** ***(If applicable)***	**7. DATE SUBMITTED *(YYMMDD)***

8. PREPARING ACTIVITY

a. NAME	**b. TELEPHONE *(Include Area Code)*** **(1) Commercial** (810) 574-8745 **(2) AUTOVON** 786-8745
c. ADDRESS *(Include Zip Code)* **Commander** **U.S. Army Tank-automotive and Armaments Command** **ATTN: AMSTA-TR-E/BLUE** **Warren, MI 48397-5000**	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** **Defense Quality and Standardization Office** **5203 Leesburg Pike, Suite 1403** **Falls Church, VA 22041-3466** **Telephone (703) 756-2340 AUTOVON 289-2340**

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:50Z
Check the source to verify that this is the current version before use.